Class-Asymmetric Robustness and Transformation-Order Effects in Lightweight Audio Deepfake Detection

Abstract

Speech deepfake detectors encounter noise and lossy coding in succession. We examine whether channel augmentation improves stressed ranking while preserving decisions at a fixed threshold. Paired MobileNetV3Small–LSTM detectors trained on 640 ASVspoof 2019 LA recordings shared initialization, source order, optimization budget, and checkpoint selection within each of two seed blocks. Development analysis showed stressed ranking gains alongside attack-specific deterioration. A locked within-release evaluation used 1,088 speaker-disjoint recordings, four conditions, and the four development-selected models. Mean stressed ROC-AUC increased from 0.8308 to 0.8875 and from 0.7498 to 0.8948. Stressed FAKE misses fell from 1,370 to 587 and from 815 to 453 out of 2,496 transformed views per block. However, stressed misses increased on A17–A19 in both blocks, clean FAKE misses increased, and clean ROC-AUC declined. Reversing noise and codec order continued to produce class-dependent decision changes. Paired speaker-cluster intervals describe uncertainty within each fitted comparison. These results support reporting aggregate ranking with class-, attack-, and source-linked errors when assessing augmentation under ordered signal processing.

Keywords. Audio forensics, channel augmentation, codec robustness, spoofing countermeasures

1 Introduction

Audio deepfake detectors are used to flag speech that may have been synthesized or voice converted. In an authentication workflow, a false alarm can divert a bona fide recording to manual review, whereas a missed spoof can leave impersonation undetected. The cost of either decision depends on the workflow and the prevalence of spoofed traffic. A detector chosen by aggregate discrimination alone can behave differently at a fixed operating point after a recording passes through ordinary signal processing.

Noise and lossy coding are particularly relevant because they can occur in either order. Speech recorded in a noisy room may be encoded before transmission, while background audio can also be introduced after decoding. These pipelines preserve the intended class of a source, but change the evidence presented to a detector. Even with identical noise and codec settings, reversing the operators can change the waveform, spectral features, ranking, and fixed-threshold decisions. Measurements across matched views can isolate the influence of the implemented signal-processing order within a recorded source.

The accepted LAVA conference manuscript benchmarks six heterogeneous detectors. The extension presented here addresses a different question through a matched intervention on one detector architecture. Within each of two initialization blocks, the training recordings, order, optimizer, numerical backend, and checkpoint rule are shared. The controlled factor is whether each training recording is presented clean or with a view sampled from a channel bank. Both fitted models are evaluated on each development recording and condition. This enables direct accounting of ranking changes, class-specific error changes, and which sources cross the same decision threshold.

The analysis asks whether aggregate robustness moves in the same direction in both blocks, which attacks and speakers bear the associated errors, and how those errors change under reversal

of a conventional codec and noise chain. The resulting evidence is a controlled comparison and a source-linked diagnostic. Sections 2–5 specify the design, and Sections 6–8 distinguish measured findings from possible explanations and threats to validity.

1.1 Operational question and scope

The action following a detector decision determines which error is costly. A system that routes suspicious clips to an analyst incurs more review when authentic audio crosses the threshold. A system used to prioritize potentially manipulated evidence has a different cost for missing a synthetic or converted clip. A balanced validation cohort therefore answers a diagnostic question about conditional errors. It does not estimate workload for a deployment with an unknown spoof prevalence. The operational quantities reported here are denominated within class and condition, so that a practitioner can consider the two errors separately.

Clean evaluation provides a useful reference point because a detector can rank bona fide and spoofed recordings well before channel processing changes their score distributions. In audio production and communications, however, noise and compression can occur in either order. It is possible to keep the source, nominal noise level and codec settings fixed while varying the order. An observed score change then identifies sensitivity to the specified composition on that recording. Comparison across two different recording lists would entangle this effect with differences in speakers, content and synthesis examples.

The training comparison addresses a complementary question. Exposure to individual channel views may help on some transformed evaluation views even when the specific compositions were absent from the training-view bank. The experiment measures this transfer within ASVspoof 2019 LA under matched training and checkpoint selection. It records whether the rank benefit and fixed-threshold behavior agree, and whether the observed differences survive inspection at the attack and speaker level. This scope places claims about ordered signal processing and model response on separate evidential footing from claims about deployment or forensic cue preservation.

1.2 Research questions

The conference study answered which available artifacts compared favorably under a common offline test and timing contract. Its clean comparison used 2,737 checksum-disjoint recordings and its stress diagnostic used a fixed 100-recording subset. Because the six artifacts had heterogeneous training histories, that comparison could not isolate the effect of changing a training-view policy. Moreover, the diagnostic retained aggregate stress results and could not identify which recordings crossed the decision boundary when the channel changed.

The journal study asks three narrower questions. First, does changing only the training-view policy improve ranking under each registered stress condition for two matched initializations? Second, which class, speaker, and attack groups account for the difference between aggregate ranking and errors at the same operating point? Third, what can a same-source reversal of noise and conventional codec order show about sensitivity to a composed channel? The last question concerns the detector's response to specified operators. Evidence for an independent forensic cue

would require another measurement.

1.3 Main Contributions

The main contributions of this study are as follows.

- **Controlled and reproducible robustness evaluation.** We compare ERM and channel augmentation for MobileNetV3Small–LSTM across two independent initialization seeds. Within each pair, the architecture, source recordings and ordering, optimization budget, frozen numerical backend, and registered checkpoint rule are shared. A locked 1,088-recording evaluation then tests the development-selected models on a speaker-disjoint partition of the same benchmark release (Sections 4 and 5).
- **Class- and attack-specific robustness characterization.** Stressed ROC-AUC improves in both blocks on development and held-out data. Development misses increase on A05 and A06, while held-out misses increase on A17–A19. Classwise and source-linked decisions localize changes that aggregate ranking conceals. Speaker-cluster intervals quantify uncertainty within each fitted comparison (Section 6).
- **Source-linked transformation-order diagnostics.** Noise-to-Opus and Opus-to-noise views of the same recording support directional score contrasts and classwise threshold transitions. Attack-level analysis and a single-seed matched-component nuisance diagnostic delimit what can be attributed to the tested processing chains (Section 6). The contrasts describe detector responses and leave the underlying forensic mechanism unresolved.

2 Related Work

2.1 Detector architectures and benchmark design

RawNet2 extracts representations directly from speech waveforms, while AASIST uses graph attention over spectro-temporal representations [1, 2]. A compact convolutional backbone offers a different compute and representation trade-off. MobileNetV3 was designed for mobile visual recognition [3]. Here, it encodes successive Mel-derived feature segments and a recurrent head aggregates them in time. Published lightweight alternatives such as DIN-CTS combine a different feature extractor with contrastive training [4]. The historical detectors provide context for this choice, and the controlled intervention uses the same backbone and head in every arm.

ASVspoof 2019 distinguishes logical-access synthetic or converted speech from physical-access replay [5]. The 2021 challenge extends benchmarking to separate logical-access, physical-access, and deepfake tasks under different conditions [6]. ASVspoof 5 scales crowdsourced collection and expands spoofing and adversarial evaluation [7]. In-the-Wild evidence shows why a benchmark split cannot by itself establish transport to collected real-world material [8]. The present development subset has speaker separation from training and includes attack IDs seen in training. In the within-release evaluation, A16 and A19 share algorithms with A04 and A06, respectively, and A17 shares a framework with A05 [5]. Those relationships constrain any interpretation of held-out attack IDs.

The multilingual MLAAD resource targets variation in synthesis systems and languages [9]. Its design provides a complementary test of transport to language and generation domains beyond this within-release comparison.

2.2 Signal transformations and robust learning

Opus is a conventional lossy codec [10]. Compression can change spectral and temporal cues through content-dependent coding decisions. Resynthesis with a learned neural codec is a different transformation family, with different artifacts. A review and case study by Maltby et al. place codec effects alongside dataset bias and other failure modes [11]. These studies motivate treating the nominal operator, signal representation, and downstream detector as separate factors.

RawBoost introduces waveform-level augmentation for spoofing countermeasures [12]. Adversarial training with frequency selection changes the perturbation rule [13], whereas group distributionally robust optimization changes the risk aggregation across predefined groups [14]. DK-CAST uses clean-to-degraded distillation [15]. SONAR studies spectral-contrastive residual representations [16]. In this comparison, the loss is ordinary binary cross-entropy, with one sampled view per source and no explicit consistency, adversarial optimization, group reweighting, or teacher signal.

2.3 Order effects and operating-point uncertainty

ADD-C tests detection under simulated conventional codec compression and packet loss [17]. Its condition coverage provides a useful comparison with a narrow within-source experiment. Bona fide cross-testing likewise shows why authentic-speech coverage matters [18]. Our conventional order reversal holds the source and noise plan fixed. It measures the effect of two specific processing orders and does not sample the distribution of real calls or search for worst-case perturbations.

Calibration concerns the interpretation of scores, while selective prediction introduces a reject action [19, 20]. Work on calibrated audio deepfake detection illustrates the importance of these distinctions in the speech domain [21]. Here the sigmoid output is denoted by p for convenience and used as an uncalibrated score. A rank improvement and an error increase at a fixed threshold are compatible. Table 1 summarizes how the nearest work differs in treatment and evaluation scope.

2.4 Channel exposure and representation choices

Work with a pretrained wav2vec 2.0 front end and data augmentation provides published evidence for a different route to addressing channel and source shift [22]. The present comparison uses a compact Mel-sequence classifier with the same representation family in both arms. This limits the treatment comparison to exposure during fitting and avoids interpreting a change of backbone, input bandwidth, or pretraining corpus as an augmentation effect. The design also avoids comparisons with a published EER obtained under a different source partition and operating protocol.

Table 1: Comparison of research designs. Different cohorts and operating rules prevent a direct performance ranking.

Study	Data or evaluation setting	Treatment or model	Relationship to the present comparison
RawBoost and F-SAT [12, 13]	ASVspoof and DeepFakeVox-HQ, respectively	Waveform augmentation and frequency-selective adversarial training	Establish robust training precedents. Neither is retrained as a matched arm here.
DK-CAST and SONAR [15, 16]	Speech spoofing and domain-shift evaluations	Distillation and spectral residuals	Intervene on learning objectives or representations. The present study intervenes on training views.
ADD-C and bona fide cross-testing [17, 18]	Simulated communication and diverse authentic speech	Codec and loss grids, and bona fide domain testing	Motivate stress coverage and authentic-speech reporting. The present study controls order on a fixed source.
Present study	Two initialization blocks sharing ASVspoof 2019 LA development	Matched lightweight detector, ordinary augmentation	Reports class, attack, speaker and per-source transitions. Transport and broader mechanism evidence remain open.

Cross-domain multilingual benchmarks such as XMAD-Bench address variation across synthesis and language domains [23]. Such benchmarks answer a different question from a source-matched transformation test. Pairing holds a source fixed across channel operations, improving interpretability of the channel contrast. It also conditions the result on the source population in the selected cohort. A cross-domain benchmark can broaden the population but introduces additional differences that must be accounted for when attributing a score shift. Both forms of evidence are relevant to deployment, and neither can replace the other.

Group-robust learning emphasizes a specified subgroup criterion and usually requires a choice of groups and weights. A standard augmentation objective instead averages the loss across sampled exposures. This distinction matters for an attack with a small number of training sources or a poorly represented channel interaction. The latter objective need not protect the worst subgroup even if it improves aggregate discrimination. The journal compares the ordinary view-policy intervention and records subgroup effects. No class-aware weighting, consistency loss, or group-robust optimizer is introduced.

3 Problem Formulation

Let x_i be source recording i, g_i its speaker, and $y_i \in \{0, 1\}$ its label, with 0 denoting REAL and 1 denoting FAKE. Let $p_\theta(Tx_i) \in [0, 1]$ be the detector score after transformation T. The decision $\widehat{y}_{i,T} = \mathbf{1}\{p_\theta(Tx_i) \geq 0.5\}$ fixes the operating point. The class-conditional risks in a fixed cohort

and condition are

$$R_0(T) = \frac{\sum_{i:y_i=0} \widehat{y}_{i,T}}{n_0}, \qquad R_1(T) = \frac{\sum_{i:y_i=1}(1 - \widehat{y}_{i,T})}{n_1}. \tag{1}$$

They correspond to false alarms and spoof misses. For a set of stress conditions, an error count adds the numerator across conditions and uses n_y times the number of conditions as the denominator. It describes risks across repeated views.

Receiver operating characteristic area under the curve (ROC-AUC) compares the ordering of FAKE and REAL scores, crediting half for ties. A monotonic score transformation can preserve ordering and change the number of scores crossing 0.5. Equal error rate (EER) is summarized at the empirical ROC point closest to equality between false-alarm and miss rates, using their mean at that point. Both metrics require support from both classes. The stress macro ROC-AUC is the equally weighted mean of ROC-AUC computed separately for the three registered stress conditions.

Let $N_i(x) = x + \nu_i$ denote insertion of the registered fixed-absolute noise for source i, and let C denote encoding and decoding with Opus. Define $T_{NC} = C \circ N_i$ and $T_{CN} = N_i \circ C$. Because C uses content-dependent analysis and lossy coding, in general $C(x + \nu_i) \neq C(x) + \nu_i$. In one case the codec processes the added noise, while in the other it does not. The contrast

$$d_i = p_\theta(T_{NC}x_i) - p_\theta(T_{CN}x_i) \tag{2}$$

measures a directional score response. A decision transition records which order alone causes an error, if either does. The contrast can reflect altered input features, score placement or cues used by the detector. An independent cue measure would be needed to interpret it as forensic-evidence retention.

3.1 Ranking and conditional risk estimate different quantities

For scores S_1 and S_0 drawn from the FAKE and REAL groups of a condition, the population form of the ranking estimand is

$$\mathrm{AUC}(T) = \Pr(S_1 > S_0) + \tfrac{1}{2}\Pr(S_1 = S_0). \tag{3}$$

The finite-sample estimate averages this comparison across class pairs [24]. Each pair contributes to ranking even when both scores lie on the same side of 0.5. Conversely, a source near 0.5 may change its decision under a small perturbation that changes almost none of the pairwise comparisons. Thus AUC and threshold errors can move in different directions. The macro stress AUC gives each of the three stress conditions equal weight. Pooling scores before ranking would also compare scores across different conditions and would estimate a different quantity.

Let $D_i^{(m)}(T)$ be the error indicator under method m. The paired contribution to a conditional error difference is $D_i^{(\mathrm{channel})}(T) - D_i^{(\mathrm{ERM})}(T)$. Its values are $+1$ for a newly introduced error, -1 for a recovered decision, and zero for agreement in error status. Summing these contributions gives the net count while retaining the ability to distinguish which sources changed in each direction. This matters when equal error counts conceal disjoint sets of failures. For an attack

group, the estimand is the miss rate conditional on that FAKE group. Ranking within a single FAKE group against itself has no binary detection interpretation.

3.2 Order effects and component diagnostics

For a frozen model, let $p_0, p_N, p_C, p_{NC}, p_{CN}$ denote scores from the clean, noise-only, codec-only and two composed inputs. The signed order contrast is $p_{NC} - p_{CN}$. A complementary descriptive contrast is

$$h_i = \tfrac{1}{2}(p_{NC} + p_{CN}) - p_N - p_C + p_0. \tag{4}$$

This contrast is zero if the two component changes add on the score scale and the two orders have the same score. Departures may also arise from nonlinear features or the sigmoid score mapping. Consequently the contrast is a diagnostic of output nonadditivity, and no inference about which speech cue was preserved follows from its magnitude. The standalone matched components are available for 9107, whereas the registered score and decision order contrasts are available for both blocks.

3.3 Decision costs and attack mixtures

Suppose a deployment has spoof prevalence π and costs c_0 and c_1 for a REAL false alarm and FAKE miss, respectively. For a fixed channel and threshold, a simple expected error cost is

$$(1 - \pi)c_0 R_0(T) + \pi c_1 R_1(T). \tag{5}$$

The prevalence and costs are not estimated here. The expression clarifies why neither class-balanced training nor an equal-weight stress macro average specifies an operational decision rule. The 0.5 rule provides a fixed point for a matched comparison and leaves calibration and cost-optimal threshold selection for a separate study.

The attack mixture also affects the observed miss rate. For represented attack IDs a with class-conditional weights w_a, $R_1(T) = \sum_a w_a R_{1,a}(T)$. Changing these weights changes the aggregate even if all attack-specific rates stay fixed. In development the mixture comes from the selected sources, and repeated views preserve their attack identity. The reported aggregate therefore pertains to this mixture. Separate attack rows enable comparison without transferring these mixture weights to another population.

3.4 Matching the intervention and defining the comparison

Let b index a training seed block, m index the two fitting policies, and $s_{ibm}(T)$ denote the FAKE score from the selected checkpoint for recording i under condition T. A between-arm score difference changes m while preserving i, b, and T. A within-model order difference instead changes T while preserving i, b, and m. The first comparison concerns the total effect of the fitting policy followed by checkpoint selection. The second concerns the detector response to a specified input manipulation. Interpreting one as the other would conflate changes in learned weights with changes in input.

A threshold transition can be analyzed without introducing a second operating point. For a REAL source, crossing from below to above 0.5 introduces a false alarm. For a FAKE source, the same crossing repairs a miss. A directional shift can therefore move the two risks in opposite directions even when its sign is common across classes. The empirical question is where each distribution lies relative to the fixed threshold and how its ordering changes relative to the other class.

4 Matched Training Methodology

4.1 Detector and objective

Audio is decoded as mono at 22.05 kHz with a three-second limit and processed using the frozen P0 input contract. Six consecutive 0.5 s windows are mapped to log-Mel power with 128 filters, a 2,048-sample Hann window and a 512-sample hop. P0 floors the power at 10^{-10}, references each segment to its maximum, clips the dynamic range to 80 dB, and uses bilinear resizing and three-channel replication to produce 224-by-224 tensors. An ImageNet-initialized MobileNetV3Small backbone encodes each segment. An LSTM with 128 units aggregates the feature sequence before a binary sigmoid head. The sealed experiment identifies the architecture and input implementation.

The backbone omits the image-classification top and uses spatial average pooling. It applies shared weights across the segment sequence. The temporal layer returns the final representation, which feeds a 64-unit rectified-linear layer and a dropout layer with rate 0.4, followed by a single sigmoid output. During partial fine-tuning, the final 20 backbone layers are eligible for updates, with batch-normalization layers frozen. During warm-up, the backbone is frozen throughout. These settings define the fitted detector and are independent of the checkpoint I/O and resume mechanism.

Both training arms minimize mean binary cross-entropy on one view per source at each visit. Read-only inspection of the sealed exports found 213 stored tensors with 1,308,401 scalar values per model. The count includes nontrainable state and does not estimate the number of variables updated in either fine-tuning stage. If V_i is the view and $q_i = p_\theta(V_i x_i)$, then

$$\mathcal{L}(\theta) = -\frac{1}{n}\sum_{i=1}^{n}[y_i \log q_i + (1 - y_i)\log(1 - q_i)]. \tag{6}$$

The empirical risk minimization (ERM) arm sets V_i to the clean view. The channel arm samples from clean, half gain, 10 dB noise, and 24 kb/s Opus. Neither ordered stress chain is a supervised training view. Training-view randomness is determined by the registered seeded procedure, while the source order and initialization are identical within each pair.

4.2 Optimization and model selection

Adam [25] is used for 840 updates over 21 epochs. Each update accumulates gradients over 16 single-source microbatches. One warm-up epoch trains the head, followed by 20 partial fine-tuning epochs with frozen batch-normalization state. The respective learning-rate stages

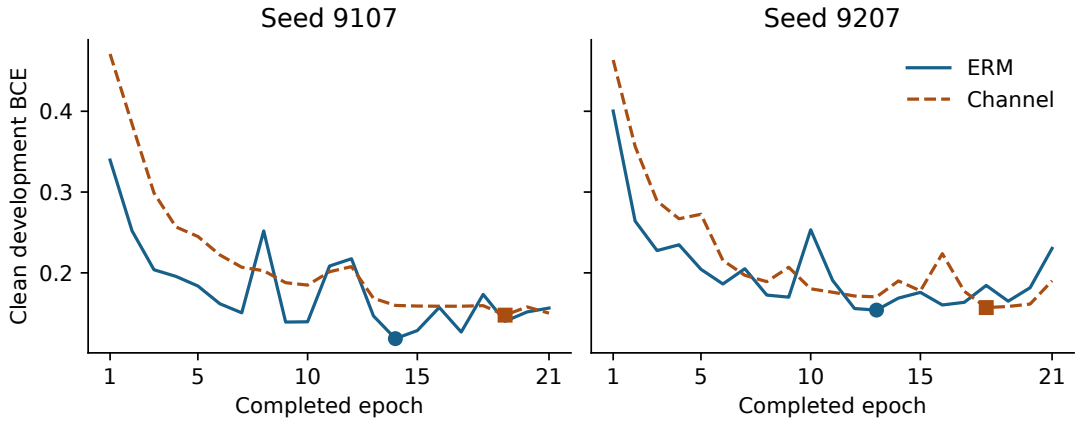


Figure 1: Recorded clean development BCE over 21 epochs for both methods and seeds. Each observation uses the same 224 development A recordings. Markers show the selected epoch for each model.

start at 0.001 and 0.0001 and use the registered linear decay. The registered CPU engine uses full-shape recomputation and a fixed numerical contract. Checkpointing and deterministic resume support reproducibility.

Validation tier A evaluates clean mixed-class development at each epoch. Tier B also measures the registered stress and REAL-only endpoints every third epoch and at completion. The minimum clean A BCE checkpoint is chosen, and tier C evaluates its complete development grid. Risk guards are evaluated as a comparison of the selected checkpoints. They do not select a different epoch or alter the fixed decision rule.

Figure 1 shows the observed clean validation trajectories. All runs completed epoch 21, although the selected checkpoints fall between epochs 13 and 19. The curves compare the same validation loss on the same cohort. Training losses reflect different view policies and are excluded from the plot.

4.3 Treatment, numerical contract and reproducibility

The two treatment arms use the same binary labels and classification objective. A change of view preserves the source identifier and training label. This keeps the number of optimizer updates and the number of source visits identical within a seed while exposing the channel arm to nuisance variation. The clean, gain, noise and codec views share the preprocessing policy. Each arm learns its own weights from its assigned views. The comparison estimates the total effect of the registered augmentation treatment and subsequent selection rule, so it includes changes in which epoch is selected.

The frozen batch-normalization policy avoids recomputing running statistics from different mixtures of training views. This controls one source of between-arm variation but also limits adaptation to the new input distribution. The warm-up and partial fine-tuning stages use the same schedule and update accounting in both arms. The optimizer accumulation unit is the committed update. Checkpoints store numerical state and the source cursor so that interrupted execution can resume without changing the registered sequence.

At inference, the selected checkpoint is restored from the sealed artifact and receives each

registered view through the same P0 feature contract. P0 combines a power floor, per-segment peak reference, dynamic-range clipping, interpolation, and channel replication. Although the peak reference reduces uniform gain sensitivity over the linear range, near-floor values, padding, clipping and interpolation can still matter. Identical preprocessing between arms makes the treatment comparison interpretable, and leaves open how the effect would change with a different feature pipeline.

The analysis records a score and class decision for every model–source–condition cell. The source and condition identity must match before paired results are computed. A missing or changed row would invalidate a paired difference. This accounting also separates the final validation of the selected checkpoint from the checkpoints used to monitor optimization. The validation record reported by a live monitor can describe a different epoch from the exported model.

4.4 Why selection is part of the intervention

Clean-loss selection imposes a common rule, while allowing the selected epoch to depend on the training policy. A comparison at the same update would answer a separate optimization question. In the deployed comparison each arm is evaluated at its own clean-selected checkpoint, with the threshold held fixed and stress scores excluded from checkpoint choice. Reporting the selected updates makes the distinction visible and prevents accidental substitution of final weights.

The identical source order prevents a difference in source scheduling from explaining an observed arm contrast within a block. Channel training consumes stochastic view choices, while the ERM arm applies the identity view. This difference is the intervention. The two transformations used in the ordered stress chains are evaluated in composition even though the composed views are absent from the training-view bank. Consequently, improvement on a composition measures transfer across views within the tested cohort and is conditional on the chosen model family and fitting budget.

5 Experimental Setup

5.1 Cohorts and conditions

The resource-bounded medium cohort includes 640 ASVspoof 2019 LA training recordings, balanced between REAL and FAKE, and attack IDs A01–A06. Development A consists of 112 REAL and 112 FAKE recordings from seven speakers. Development B adds 144 REAL-only recordings from nine speakers. The registered audit checks disjoint training and development speaker IDs. The same development sources are used for the two independently initialized blocks, so the blocks measure sensitivity to training, rather than to a new test population.

The seven scored views per model are clean, two gain controls, Noise5, Opus64, and the two reversed chains. The stress endpoint uses Noise5 and the chains. In the chains, a fixed-absolute 10 dB noise plan and the 24 kb/s Opus encoder are shared across order. Noise5 and Opus64 are separate conditions. The matched standalone Noise10 and Opus24 diagnostic was performed for seed 9107 only. Table 2 summarizes the fixed design.

Table 2: Registered prospective development experiment.

Factor	Specification
Training	640 sources, 320 per class
Development	A 224 mixed-class, B 144 REAL
Blocks	Matched seeds 9107 and 9207
Architecture	MobileNetV3Small with LSTM128
Budget	21 epochs, 840 updates per arm
Selection	Minimum A clean BCE
Operating point	FAKE when score is at least 0.5
Outputs	2,576 selected-model scores per arm

5.2 Paired uncertainty and efficiency

For each seed, differences are channel minus ERM. Percentile intervals use 10,000 paired speaker-cluster bootstrap draws with seed 91109. A and B are sampled separately. Within a sampled speaker, all sources and their views retain their paired model scores. This preserves source and speaker dependence and avoids treating transformed views as independent trials. Mean and range across the two blocks are descriptive. They do not estimate a population confidence interval over random initializations.

The four runs used a Windows CPU environment. Optimizer and validation durations come from run logs and may include repeated work after interruption. The registered evaluation policy preserves the 0.5 threshold and excludes calibration from model selection.

5.3 Design controls and support

Development A and B play different roles. A supplies both classes for ROC-AUC, EER and fixed-threshold risks. B adds a second, REAL-only speaker stratum for false-alarm sensitivity. Pooling B into A would change the implicit weighting of speakers and the set of REAL–FAKE pairs. The two strata are retained separately. Counts in the mixed-class stratum are supported by 112 original recordings per class. The three stress views contribute 336 outcomes per class but still derive from those same recordings.

Within each seed block, both fitted models see the same registered development sources and transformed waveforms. Between seeds, the independent initializations vary the fitted functions, but the evaluation cohort stays fixed. Thus there are two kinds of pairing. Source pairing controls the input to a between-method comparison, and seed blocking controls initialization and source order during training. Neither kind of pairing gives two independent samples of the speaker population. All attacks A01–A06 are represented in training. The attack rows therefore show differences across represented attack IDs under stress.

The order experiment inserts the same fixed-absolute noise realization before or after the codec, with the same nominal clean-referenced signal-to-noise ratio. The achieved characteristics after compression need not be equal. Treating achieved SNR or bandwidth as equal would erase part of the intervention. The controls support comparison of those specific pipelines and do not simulate packet loss, device reverberation, proprietary enhancement or a complete communications stack.

5.4 Interpretation of the resampling intervals

Cluster resampling keeps all observations from each sampled speaker together [26]. In each replicate the same speaker weights apply to ERM and channel within a stratum. A speaker drawn more than once contributes the corresponding repeated cluster weight. The analysis recalculates the contrast under those weights and takes the percentile limits over valid draws. Repeated views of a source and paired arm predictions therefore stay linked. For rank metrics, the class support requirement is evaluated within the resampling design. The evidence records the valid draw count so that interval interpretation can account for any undefined draws.

Seven and nine speakers provide limited empirical support for percentile intervals. Large changes in one speaker can influence the interval and can also affect which attack cases are represented in a resample. The intervals condition on the fitted models and do not incorporate a hypothetical new training seed. The differences across seeds are summarized descriptively, and the secondary attack and speaker comparisons are used to localize effects rather than to claim a family of discoveries.

5.5 Evidence boundaries

The prospective comparison includes the registered score orientation, 0.5 operating rule, fixed budget, clean-loss checkpoint selection and stress conditions. The paired records define the unit for contrasting methods. The separate Noise10 and Opus24 runs for 9107 were added as diagnostics on the same fitted models and development sources. They help characterize departures from additive output behavior, with no independent new-cohort replication. The interpretation of component nonadditivity and nuisance adjustment therefore stays distinct from that of the prospective augmentation comparison.

5.6 Locked Held-Out Evaluation

The primary held-out subset comprises 1,088 recordings from 67 speakers in the official ASVspoof 2019 LA countermeasure evaluation partition [5]. A deterministic hash ranking and speaker round-robin selection with seed 91109 yields 256 REAL recordings and 64 FAKE recordings for each attack A07 through A19. The audit reconciled all 71,237 official trials with the archive and found no shared source IDs, speaker IDs, or audio SHA-256 digests between the selected subset and the registered training, development, and calibration manifests. Complete generator and original-utterance lineage remains unverified.

The lock specifies four development-selected checkpoints and four conditions per source: clean, Noise5, Noise10 followed by Opus24, and Opus24 followed by Noise10. All 17,408 predictions were completed. The sigmoid score is oriented toward FAKE and scores of at least 0.5 are assigned to FAKE. Macro stressed ROC-AUC averages the three stress-condition AUCs. Clean and stressed error rates use class-conditional denominators, and the stressed denominator counts three correlated views per source. Held-out contrasts retain within-seed source pairing and use the same speaker-bootstrap settings on the held-out speakers. Development A and B refer only to the earlier development strata.

All source–condition–model cells were checked against the locked identities. Per-source score

Table 3: Development results. Stress counts use repeated views. Timing is from CPU logs.

Development endpoint	9107 ERM	9107 Channel	9207 ERM	9207 Channel
Selected update	560	760	520	720
Clean A ROC-AUC	0.9872	0.9814	0.9851	0.9809
Clean A EER (percent)	4.46	4.91	5.36	5.36
A macro stressed ROC-AUC	0.6715	0.8026	0.5946	0.7839
A clean REAL false alarms	3/112	3/112	4/112	4/112
A clean FAKE misses	6/112	9/112	9/112	9/112
A stressed REAL false alarms	54/336	45/336	167/336	70/336
A stressed FAKE misses	222/336	155/336	143/336	134/336
B clean REAL false alarms	19/144	31/144	26/144	20/144
Optimizer minutes	42.6	82.9	41.5	82.4
Validation minutes	40.2	39.1	41.3	39.5

shards retain complete model and condition sets, and their hashes were verified before analysis. The registered metrics and uncertainty outputs were recomputed from the committed predictions. The selected sources, models, transformations, and threshold were fixed before scoring. The attack-balanced subset has different support from development A and B, so differences in effect magnitude across partitions are descriptive.

6 Results

6.1 Ranking and operating-point errors

Table 3 reports selected checkpoints, clean and stressed performance, and CPU work. Figure 2 connects the two methods within each seed block. Stress macro ROC-AUC increases in both blocks, by 0.1312 and 0.1893 AUC units. Stress FAKE misses decline by 67 and 9 of the 336 views, respectively. Both selected channel checkpoints have slightly lower clean A ROC-AUC, with equal or slightly worse clean A EER.

Clean A REAL false-alarm counts match within each seed. Clean A FAKE misses rise from 6 to 9 in 9107 and remain 9 in 9207. Clean B REAL false alarms rise from 19 to 31 in 9107 but decline from 26 to 20 in 9207. Figure 3 contrasts those changes with the stressed A rates. Table 4 gives paired intervals and descriptive seed ranges. The stress FAKE miss interval excludes zero for 9107 and includes zero for 9207.

6.2 Attack and speaker heterogeneity

Table 6 and Fig. 4 show large reductions on A01, A03 and A04 in both blocks. A02 worsens in 9207. A05 and A06 worsen in both blocks. Each of the latter attacks has 18 sources and 54 stressed views, and their 9107 intervals include zero. The score-based guard comparison flags A clean FAKE misses, B clean REAL false alarms and A06 stress misses in 9107, and A05/A06 stress misses in 9207.

Table 7 localizes the A05 excess to the ordered chains, since Noise5 is unchanged and clean errors are zero. For A06, clean misses are already present. In 9107 the largest stress increase comes from the Noise-to-Opus chain. In 9207 the largest increase comes from Opus-to-Noise. In

Table 4: Channel minus ERM. Brackets give within-seed 95 percent speaker-cluster intervals. Error differences are percentage points.

Endpoint	Seed 9107	Seed 9207	Mean	Range
A macro stress ROC-AUC	+0.1312 [+0.0207, +0.2387]	+0.1893 [+0.0932, +0.2684]	+0.1602	+0.1312 to +0.1893
A clean REAL false alarms	+0.00 [-2.68, +2.68]	+0.00 [-2.68, +2.68]	+0.00	+0.00 to +0.00
A clean FAKE misses	+2.68 [+0.00, +6.25]	+0.00 [-3.57, +4.46]	+1.34	+0.00 to +2.68
A stressed REAL false alarms	-2.68 [-8.63, +4.76]	-28.87 [-37.20, -19.05]	-15.77	-28.87 to -2.68
A stressed FAKE misses	-19.94 [-25.31, -14.58]	-2.68 [-10.42, +3.57]	-11.31	-19.94 to -2.68
B clean REAL false alarms	+8.33 [+1.39, +17.36]	-4.17 [-6.94, -0.69]	+2.08	-4.17 to +8.33
A05 stressed FAKE misses	+3.70 [-3.70, +12.70]	+16.67 [+3.70, +29.82]	+10.19	+3.70 to +16.67
A06 stressed FAKE misses	+7.41 [-6.67, +22.92]	+27.78 [+11.76, +43.75]	+17.59	+7.41 to +27.78

Table 5: Development A by condition. Each cell gives ROC-AUC / REAL false alarms / FAKE misses. Each error count has 112 sources of its class.

Condition	9107 ERM	9107 Channel	9207 ERM	9207 Channel
Clean	0.9872 / 3 / 6	0.9814 / 3 / 9	0.9851 / 4 / 9	0.9809 / 4 / 9
Half gain	0.9812 / 0 / 15	0.9818 / 0 / 14	0.9758 / 0 / 16	0.9820 / 0 / 15
Quarter gain	0.9237 / 7 / 20	0.9576 / 0 / 17	0.9278 / 9 / 20	0.9590 / 0 / 19
Noise5	0.6567 / 1 / 99	0.7793 / 8 / 64	0.5449 / 22 / 77	0.7579 / 13 / 56
Opus64	0.9884 / 6 / 4	0.9819 / 4 / 6	0.9861 / 9 / 5	0.9839 / 5 / 8
Noise to Opus	0.6763 / 48 / 38	0.8108 / 28 / 37	0.6408 / 104 / 13	0.7981 / 44 / 26
Opus to Noise	0.6814 / 5 / 85	0.8178 / 9 / 54	0.5981 / 41 / 53	0.7958 / 13 / 52

9207, at least one stressed decision worsens for 8 of 18 A05 and 10 of 18 A06 sources, with no stress recovery in either subgroup. This shows changes across multiple recordings in the cohort without implying that the attacks' generating mechanisms have been identified.

Figure 5 shows that two B speakers contribute 10 of the 12 net additional false alarms in 9107. For 9207, reductions occur across several speakers. B contains only REAL recordings, and the counts are interpreted as false alarms with no B ROC-AUC or EER.

6.3 Processing order and matched components

Among 112 A REAL sources, the Noise-to-Opus-only false-alarm count changes from 43 to 19 in 9107 and from 63 to 31 in 9207. The opposite order has no exclusive REAL false alarms in A. Among 112 A FAKE sources, Opus-to-Noise alone causes 48 versus 17 misses in 9107 and 40 versus 26 in 9207. The Noise-to-Opus-only FAKE counts are 1 versus 0 and 0 versus 0. Both seeds therefore show decision asymmetry by class and order, with different magnitudes. Figure 6 shows the paired score contrasts before the threshold is applied.

A retrospective 9107 diagnostic adds 1,472 predictions for standalone Noise10 and Opus24. Matched waveforms reconstruct both registered chains and selected anchor scores replay without observed drift. A leave-one-speaker-out nuisance regression yields directional class-discrimination

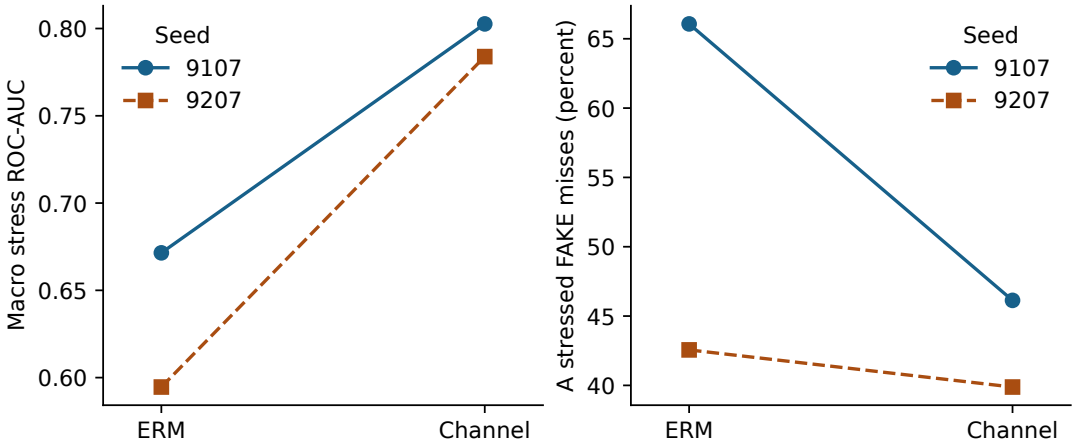


Figure 2: Development results by matched initialization. The stress FAKE miss denominator is 336 views from 112 FAKE sources in each arm. Lines connect the two treatment arms within a seed.

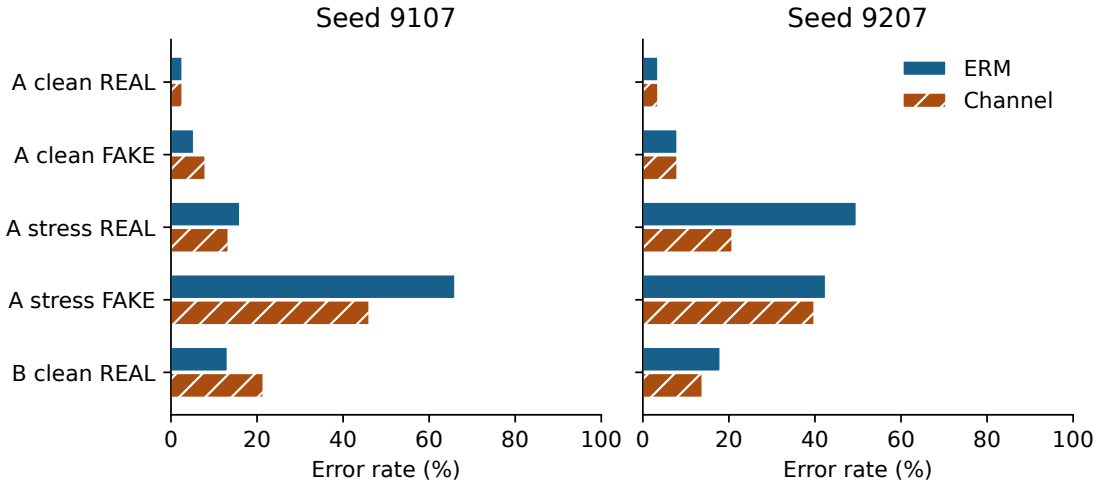


Figure 3: Fixed-threshold error rates by development stratum, class, and seed. REAL errors are false alarms and FAKE errors are misses. Clean A has 112 recordings per class, stress A has 336 views from 112 recordings per class, and clean B has 144 REAL recordings.

AUCs of 0.538 for ERM and 0.432 for channel on the signed order score. This measures a residual score contrast on the development cohort. The additional single-component analysis was not repeated in 9207.

6.4 Compute

The optimizer and validation durations in Table 3 show the recorded workload for this medium cohort. The channel arm used more optimizer wall time in both blocks, while validation durations were similar within each block. The recorded execution may include repeated work following interruption. These logs do not separate every audio transformation or data-cache effect and cannot be extrapolated to a foundation-scale budget.

Table 6: Attack-specific stressed FAKE misses over Noise5 and both ordered chains.

Attack	9107 ERM	9107 Channel	9207 ERM	9207 Channel
A01	21/57	2/57	13/57	1/57
A02	47/57	46/57	27/57	45/57
A03	38/57	8/57	30/57	7/57
A04	27/57	4/57	19/57	3/57
A05	48/54	50/54	38/54	47/54
A06	41/54	45/54	16/54	31/54

Table 7: A05 and A06 source-level diagnostics. Each condition has 18 sources. Error cells are ERM / channel. Harm and recovery can overlap across views.

Seed / attack	Clean	Noise5	Noise to codec	Codec to noise	All stress	Harmed / rec.
9107 / A05	0 / 0	17 / 17	14 / 16	17 / 17	14 / 16	3 / 1
9107 / A06	3 / 6	18 / 16	7 / 12	16 / 17	6 / 12	7 / 2
9207 / A05	0 / 0	17 / 17	7 / 13	14 / 17	7 / 13	8 / 0
9207 / A06	5 / 7	9 / 12	0 / 5	7 / 14	0 / 5	10 / 0

6.5 Reading the paired estimates

Table 5 separates the components of the macro stress endpoint. The channel arm improves AUC under each registered stress condition in both seeds. Figure 7 shows the full ROC curves underlying those AUCs. Fixed-threshold changes remain heterogeneous. Under Noise5, FAKE misses fall from 99 to 64 in 9107 and from 77 to 56 in 9207. The corresponding REAL false-alarm counts rise from 1 to 8 in 9107 and fall from 22 to 13 in 9207.

The larger 9207 REAL benefit under the two compositions is visible in the per-condition counts. Noise-to-Opus false alarms fall from 104 to 44, while its FAKE misses increase from 13 to 26. In the reverse order, REAL false alarms fall from 41 to 13, while FAKE misses change from 53 to 52. Each count has denominator 112. The baseline's high REAL false-alarm count under Noise-to-Opus provides room for a large reduction at this operating point. It would be misleading to describe that reduction alone as an across-class benefit for the condition.

The gain controls and Opus64 provide context for the stress results. Under quarter gain, the channel arm has fewer false alarms and higher AUC in both blocks. Under standalone Opus64, the channel arm has fewer false alarms but more FAKE misses in both blocks. The 64 kb/s control uses a different rate from the 24 kb/s compositions. It is reported as a distinct condition, and any comparison with the chains also changes the codec setting and presence of noise. The matched 24 kb/s diagnostic is the appropriate standalone component comparison for 9107.

The larger increase in stressed macro AUC occurs for 9207, but the larger decline in stressed FAKE misses occurs for 9107. The two statements are compatible because the estimands depend on different aspects of the score distribution. The A stress miss differences are about -19.94 and -2.68 percentage points, respectively. The mean across seed blocks is about -11.31 points and describes these two fitted comparisons. It is neither a pooled estimate from independent cohorts nor an interval over new initializations.

In 9207, reductions in other attacks offset 24 additional A05 and A06 stressed misses. Those two attacks contribute six additional misses in 9107. These differences count transformed views.

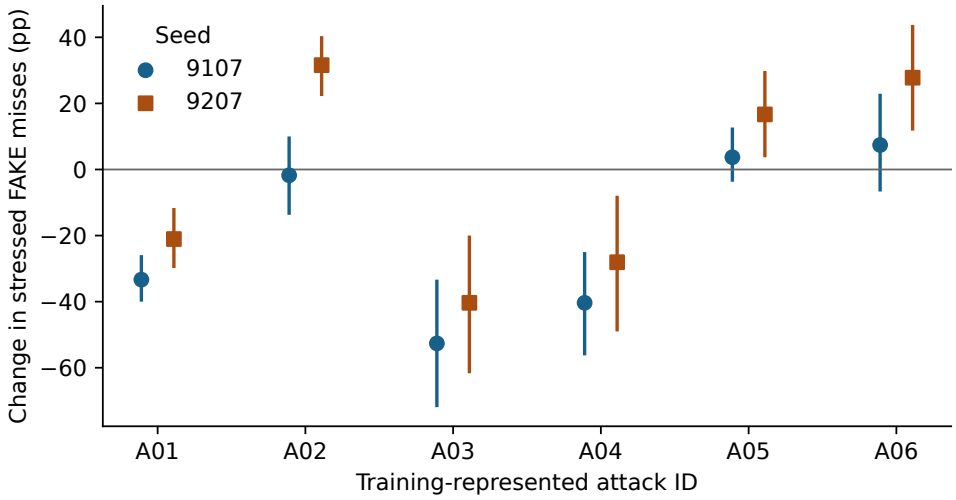


Figure 4: Stressed FAKE miss changes by attack and seed. Positive differences indicate more misses. Intervals resample speakers and are descriptive. Denominators are 57 views for A01–A04 and 54 for A05–A06.

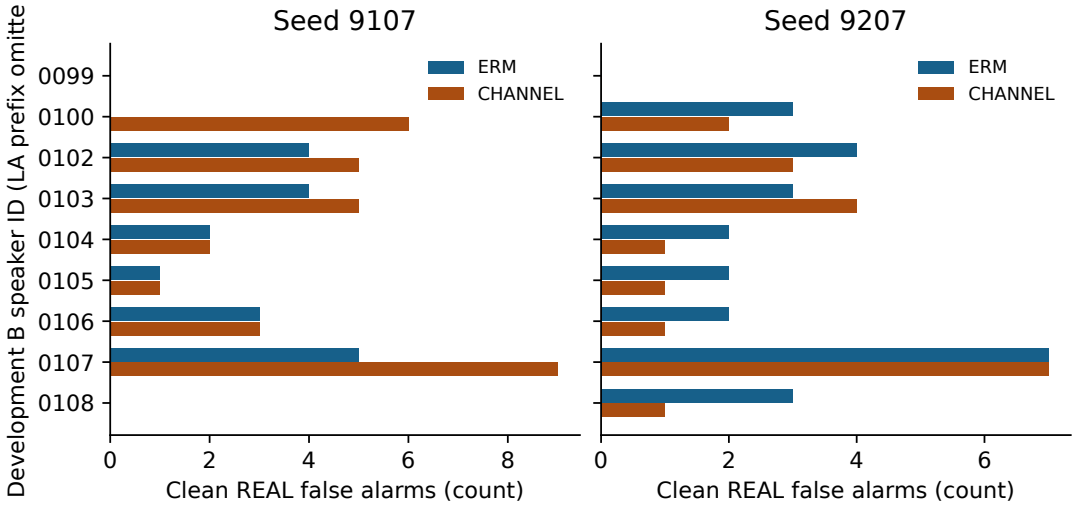


Figure 5: Clean REAL false alarms by speaker on development B. Each of nine speakers contributes 16 recordings, for 144 recordings per arm and seed.

Table 7 reports how many distinct A05 and A06 recordings gain or lose a correct decision in at least one stress condition. Figure 8 extends the paired accounting to both classes across A.

The mean difference in stressed ranking is positive in each observed seed. This supports a repeatable direction under the two registered initializations. It provides less information about the stability of threshold behavior, which varies by endpoint. The B stratum in particular would lead to opposite conclusions if only one block were reported. Reporting that reversal alongside the conditional speaker intervals prevents the aggregate score gain from being interpreted as universal risk reduction.

6.6 Resampling support for the primary contrasts

For stressed ROC-AUC, stressed FAKE misses, and B clean REAL false alarms, each seed has 10,000 valid and zero invalid speaker-cluster draws. The 9207 stress AUC contrast of 0.1893 has

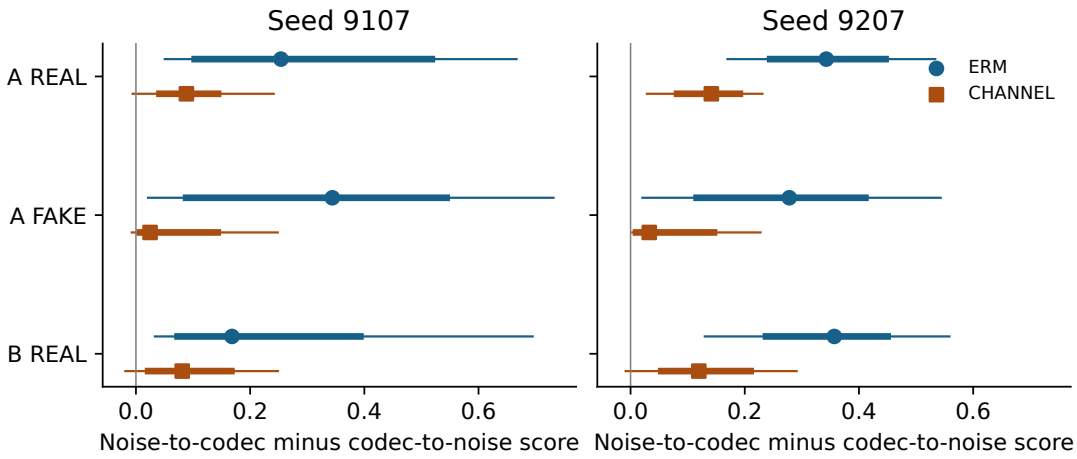


Figure 6: Directional score contrasts for the same recording in opposite processing orders. Points are medians, thick segments span quartiles, and thin segments span the 10th to 90th percentiles. Each A class has 112 sources and B has 144 REAL sources. Segments describe score distributions.

Table 8: Selected-model results on the locked held-out subset. Stress error counts pool three views per source. Macro stress ROC-AUC averages the three condition-specific AUCs.

Endpoint	9107 ERM	9107 Channel	9207 ERM	9207 Channel
Macro stress ROC-AUC	0.8308	0.8875	0.7498	0.8948
Stress FAKE miss	1370/2496	587/2496	815/2496	453/2496
Stress REAL false alarm	80/768	81/768	322/768	136/768
Clean REAL false alarm	25/256	19/256	22/256	16/256
Clean FAKE miss	125/832	129/832	127/832	147/832

a percentile interval of [0.0932, 0.2684], while its stress FAKE miss contrast of −2.68 percentage points has interval [−10.42, +3.57]. Figure 9 shows the differing directions and uncertainty across ranking and error endpoints. The B false-alarm intervals lie on opposite sides of zero. These intervals describe uncertainty from resampling speakers within each fitted comparison.

6.7 Development-to-Held-Out Comparison

The locked evaluation completed 17,408 predictions for the same four selected models. Table 8 reports aggregate results and Table 9 separates the conditions. Macro stressed ROC-AUC increased in both seed blocks, from 0.8308 to 0.8875 in 9107 and from 0.7498 to 0.8948 in 9207. The corresponding paired speaker-cluster intervals for channel minus ERM were [0.0296, 0.0842] and [0.1158, 0.1752]. These positive differences were smaller than the development A contrasts. Figure 10 compares the predefined endpoints across partitions. Figure 11 shows the score-level ranking under each held-out condition.

Stressed FAKE misses decreased from 1,370 to 587 in 9107 and from 815 to 453 in 9207, each out of 2,496 correlated views from 832 FAKE sources. The rate differences were −31.37 percentage points [−33.41, −29.36] and −14.50 points [−17.38, −11.67]. Stress REAL false alarms changed from 80 to 81 and from 322 to 136, out of 768 views from 256 sources. The respective intervals were [−3.23, +3.76] and [−29.07, −19.03] percentage points. All registered endpoint and attack intervals used 10,000 valid draws with zero invalid draws. These intervals

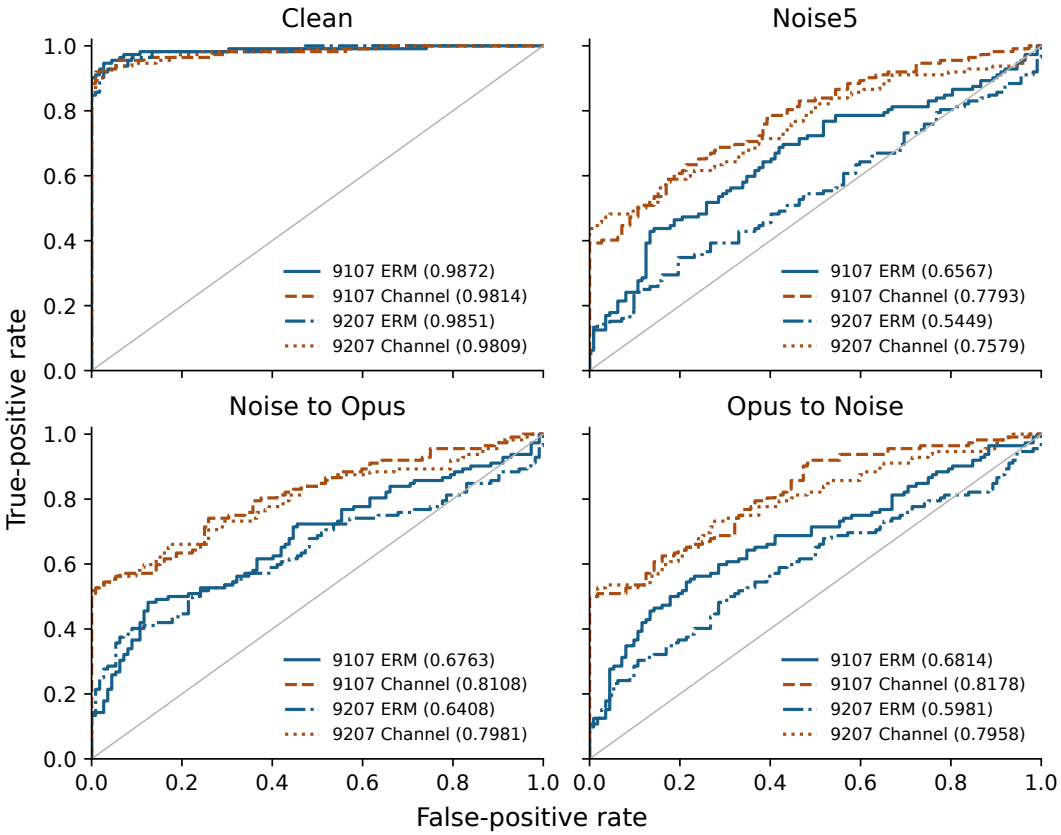


Figure 7: Empirical ROC curves from the four selected models on development A. Each panel ranks the same 112 REAL and 112 FAKE recordings under one condition. Legends give the seed, method, and recomputed AUC. The diagonal denotes chance ranking.

describe speaker resampling within a fitted seed pair.

Clean performance showed a different balance. REAL false alarms decreased by six recordings in each block, from 25 to 19 and from 22 to 16 out of 256. Clean FAKE misses increased from 125 to 129 and from 127 to 147 out of 832, while clean ROC-AUC declined from 0.9534 to 0.9436 and from 0.9507 to 0.9410. In 9207 the clean FAKE miss increase was 2.40 percentage points [1.15, 3.86]. Thus the held-out ranking improvement under stress coexisted with a clean FAKE sensitivity loss. The development B false-alarm reversal did not recur as a clean held-out reversal: both held-out pairs had fewer clean REAL false alarms. B and held-out have different speaker and source support.

6.8 Held-Out Attack and Order Effects

Figure 12 shows the condition-specific changes for A07–A19. Stress misses declined on A07–A16 in both blocks. They increased on A17, A18 and A19 in both blocks, by 10, 11 and four views in 9107 and by 43, 43 and 56 views in 9207. Each attack contributes 64 sources and 192 stressed views. The 9107 intervals for these three increases include zero, while the 9207 intervals are [15.90, 29.03], [14.12, 30.81] and [20.63, 37.70] percentage points. These descriptive attack contrasts expose adverse subgroups alongside the pooled improvement. The development A05/A06 observations and held-out A17–A19 observations refer to different attack sets.

Source-linked order asymmetry persisted. In 9107, REAL sources wrong only under noise-

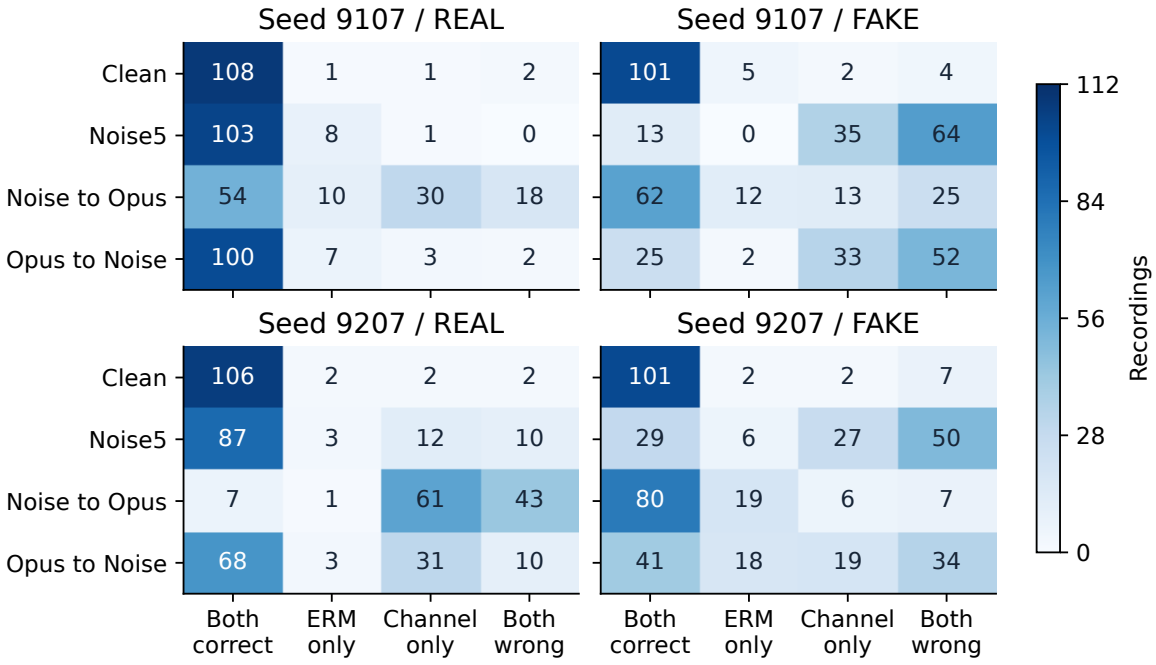


Figure 8: Within-source correctness under ERM and channel augmentation at threshold 0.5 on development A. Each row contains 112 recordings of the indicated class. The same recordings contribute to each condition row. The two middle columns identify sources correctly classified by only one method.

Table 9: Condition-specific held-out performance. Each cell gives ROC-AUC / REAL false alarms / FAKE misses. There are 256 REAL and 832 FAKE recordings per condition.

Condition	9107 ERM	9107 Channel	9207 ERM	9207 Channel
Clean	0.9534 / 25 / 125	0.9436 / 19 / 129	0.9507 / 22 / 127	0.9410 / 16 / 147
Noise5	0.8053 / 0 / 701	0.8655 / 12 / 259	0.6815 / 28 / 492	0.8765 / 25 / 193
Noise to Opus	0.8447 / 76 / 143	0.8991 / 53 / 143	0.8196 / 226 / 38	0.9044 / 89 / 93
Opus to noise	0.8424 / 4 / 526	0.8979 / 16 / 185	0.7483 / 68 / 285	0.9034 / 22 / 167

to-Opus numbered 72 for ERM and 37 for channel, with zero exclusive errors in the reverse direction. In 9207 the corresponding counts were 158 and 68, while exclusive reverse-order errors numbered zero and one. Among FAKE sources, reverse-order-only misses numbered 384 and 43 in 9107 and 247 and 74 in 9207. The opposite exclusive FAKE miss counts were one and one in 9107, and zero and zero in 9207. Each model uses the same 256 REAL and 832 FAKE sources in the order comparison.

The score contrasts in Fig. 13 also remained directional, with lower FAKE-class medians after channel training. Median FAKE contrasts changed from 0.3454 to 0.0063 in 9107 and from 0.2419 to 0.0076 in 9207. Smaller score contrasts alone do not imply a uniformly better decision rule, as shown by the attack-specific misses. Speaker-linked outputs retain the distribution of order errors. For example, ERM 9107's reverse-minus-forward FAKE error differences were 16 among 21 sources for LA_0045 and 13 among 21 for LA_0046. These speaker counts describe concentration within the selected corpus and support no demographic inference.

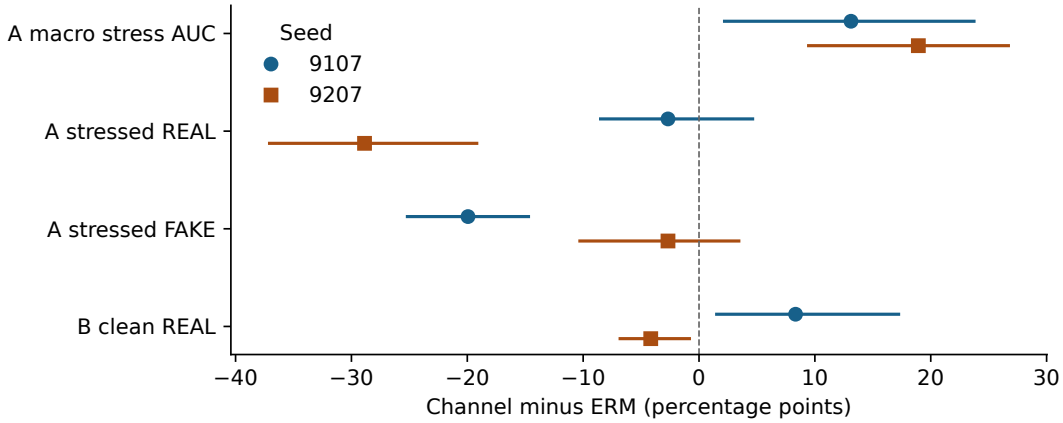


Figure 9: Channel-minus-ERM contrasts and within-seed 95% paired speaker-cluster percentile intervals. AUC differences are multiplied by 100 to share a percentage-point axis with the error differences. Positive values indicate better ranking for AUC and more errors for error rates.

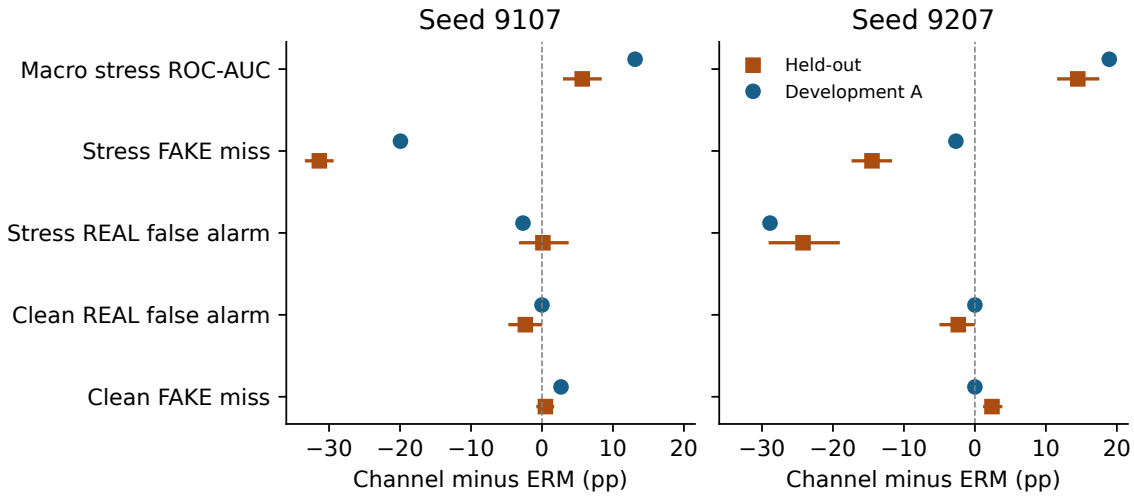


Figure 10: Channel-minus-ERM contrasts on development A and the locked held-out subset, separately by seed. Held-out segments show 95% paired speaker-cluster percentile intervals. AUC differences are multiplied by 100. Positive differences indicate better AUC or higher error rates. Cohorts and attack populations differ across partitions.

7 Discussion

On development A, aggregate stressed ranking improves in both observed blocks, with a descriptive mean difference of approximately 0.1602 ROC-AUC units. A cleaner separation of many stressed sources can raise the average even if particular attack groups shift below the chosen threshold. The measured changes in attacks A01, A03 and A04 and the worsening on A05 and A06 show why aggregate ranking is insufficient for selecting an operating point. The latter two attacks are represented in training. Their deterioration reflects a within-benchmark robustness gap under the tested intervention.

The error effect depends on initialization. The A stressed FAKE miss reduction is larger for 9107, and the 9207 cluster interval includes zero. Clean B false alarms move in opposite directions, with the 9107 increase concentrated in two speakers. Changing the score distribution relative to 0.5 could contribute to these changes, even if discrimination improves. The results

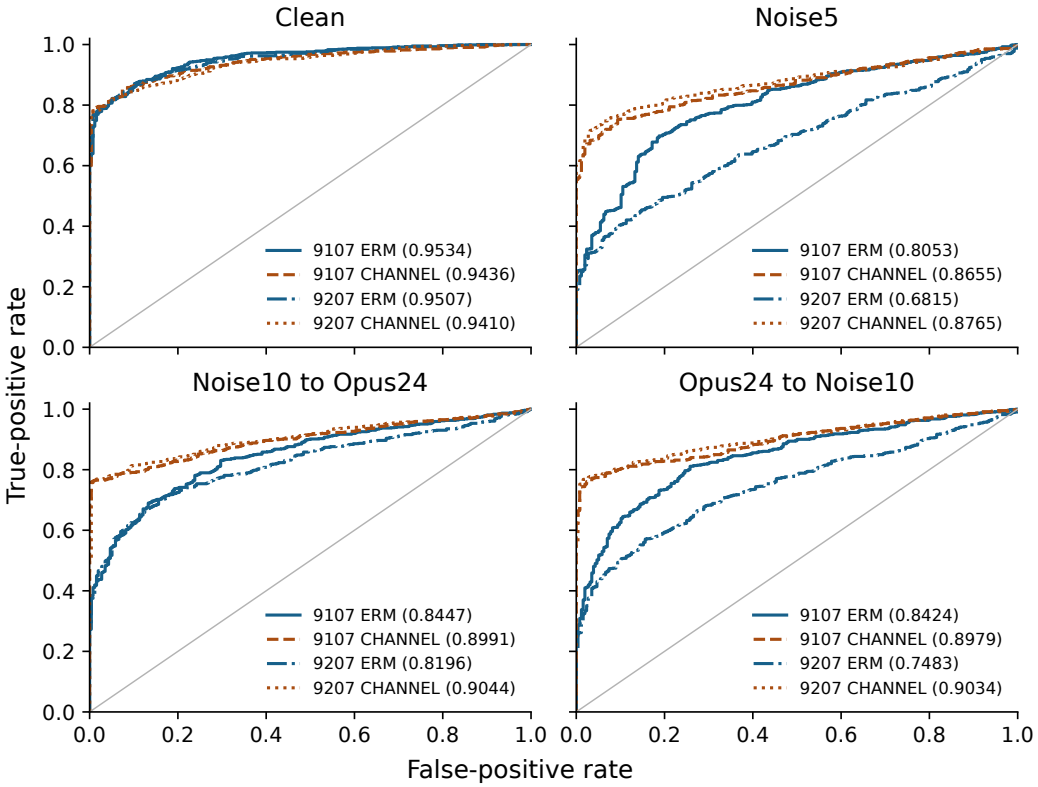


Figure 11: Empirical held-out ROC curves for four selected models. Each panel contains 256 REAL and 832 FAKE recordings, evaluated under the named condition. Legends give seed, method, and AUC. The same sources recur across conditions.

describe decisions under the registered threshold and do not evaluate threshold recalibration.

For many REAL sources, Noise-to-Opus is the only erroneous order. For FAKE sources, the reverse chain is often the only erroneous order. An upward shift in FAKE scores under one order can produce both patterns. The retrospective matched-component diagnostic for 9107 finds that nuisance adjustment does not establish an independent forensic target. Nonlinear codec response, preprocessing, and detector sensitivity may all contribute. Reproducing the direction of decision asymmetry in 9207 does not independently reproduce the nuisance analysis.

In practical evaluation, reporting both threshold-specific risks and ranking separates several failure modes that may have different costs. Speaker concentration also affects how a review workload might change, although no field workload or utility was measured. Conventional Opus order reversal, class-wise counts and a matched training intervention provide a focused complement to broader robustness benchmarks and codec-adaptation methods.

7.1 Aggregate improvement and its distribution

The matched comparison makes training-view policy the controlled factor, but does not isolate a single cause inside the fitted network. The channel model can learn features that distinguish more stress-processed pairs, move score distributions relative to the fixed decision boundary, or do both. Clean BCE selection may choose a different epoch in each arm because the fitted score distributions are different. The observed contrast therefore includes the interaction of augmentation with the same selection rule. It cannot determine which intermediate representation

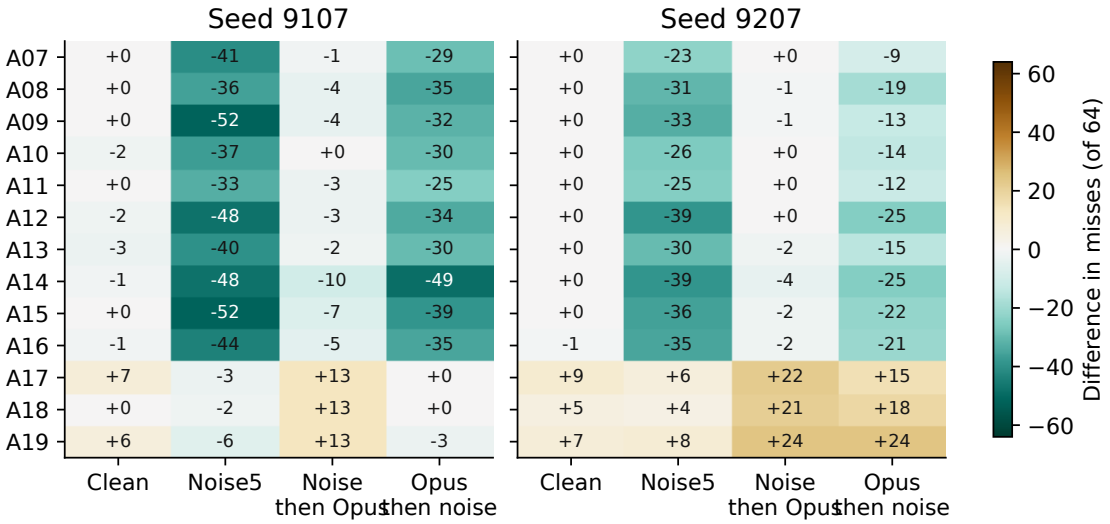


Figure 12: Held-out channel-minus-ERM FAKE miss counts by attack, condition and seed. Each cell compares the same 64 sources. Positive counts indicate deterioration and negative counts indicate fewer misses. The full numerical supplement retains both arm counts and stressed attack intervals.

would be responsible, or how an alternate selection rule would change outcomes.

The adverse results for A05 and A06 matter in two distinct ways. First, these groups already contribute repeated stress errors before augmentation. Their denominators are small, so an individual source can affect several condition counts. Second, the registered model comparisons show additional errors distributed across several sources, especially in the 9207 block. The per-condition breakdown distinguishes an attack already difficult under Noise5 from additional failures concentrated in the ordered compositions. This supports a localization claim about the tested inputs. It does not show which characteristics of their synthesis pipelines caused the effect.

REAL false alarms can shift if a score displacement moves bona fide examples near 0.5 across the threshold. In B the two blocks move in opposite directions, and part of the 9107 increase concentrates in a small number of speakers. Bona fide cross-testing work makes the broader case for assessing bona fide diversity [18]. Our speaker analysis addresses a narrower issue inside one fixed cohort. A different bona fide corpus, language, noise mixture or deployment prior could change both the baseline rates and the direction of the intervention effect.

7.2 Signal-chain and engineering implications

Published communication benchmarks such as ADD-C vary codec and packet-loss conditions [17]. Their breadth is useful for establishing stress coverage. The same-source order contrast here trades breadth for experimental control. It compares two explicitly defined pipelines at fixed settings and records exactly which order makes a particular decision fail. The association can motivate a follow-up test with actual transmission and varied gain controls. It cannot stand in for that test because the present experiment omits network loss and device processing.

Where a review decision is threshold based, a deployment report should specify the order and settings of the signal-processing chain and give error counts by class. A rank statistic can summarize detector ordering, and an error transition table can show whether a system has shifted

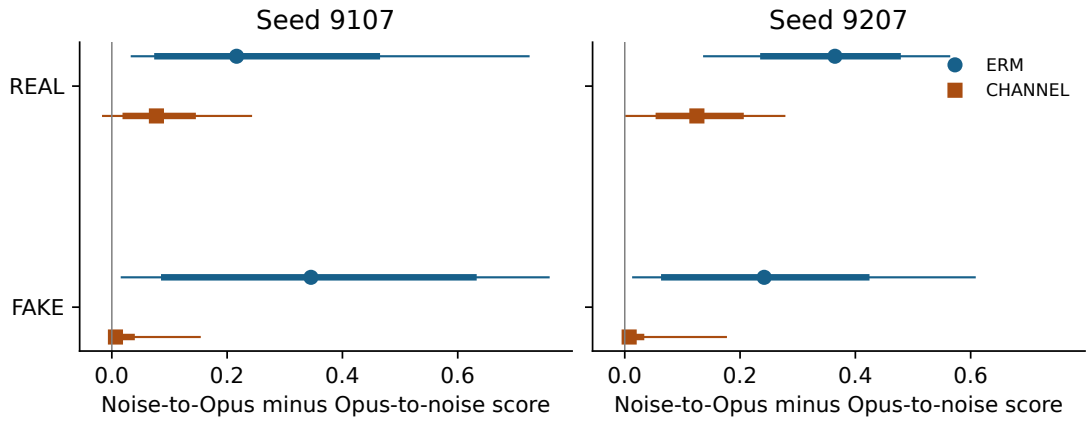


Figure 13: Held-out same-source FAKE score contrasts between noise-to-Opus and Opus-to-noise. Points are medians, thick segments span quartiles, and thin segments span the 10th to 90th percentiles. REAL and FAKE groups contain 256 and 832 sources. These are distribution summaries of paired scores.

errors to a different group. The latter is valuable even if total counts tie. For example, two methods can flag the same number of bona fide sources but require review of different recordings. No application-specific cost or calibration was measured, so a cost-optimal operating rule remains a separate problem.

The present controlled intervention uses ordinary BCE and one view per source. Its scientific contribution is the error accounting and the evidence that the observed rank benefit coexists with subgroup deterioration. Extending the claim would require a locked independent partition, then genuinely new corpora or models to measure broader transport. Confirmation of a forensic mechanism would additionally require a cue definition and a measurement that can vary independently of the detector output.

7.3 From a validation metric to a validation workflow

For an engineer deciding whether to adopt augmentation, the condition table distinguishes several cases. An AUC increase with reductions in both error types shows improvement at both levels measured. An AUC increase with fewer false alarms and more misses indicates that a rank gain and an operating-point trade-off coexist. A near-tie in counts can still conceal a change in the individual sources that require attention. The present results exhibit these behaviors under different conditions. A validation workflow benefits from retaining source identity through the paired analysis.

Nominal condition names can also obscure the processing intervention. A label that names noise and Opus omits whether the noise is present during coding. The operator descriptions here fix that detail and define how the noise realization is shared. This isolates sensitivity to the specified order without assuming either chain represents all communication paths. It also motivates reporting codec rates alongside order. Opus64 has different settings and support from the 24 kb/s compositions, and their contrast cannot isolate bitrate. The present evidence pertains to a classifier after specified transformations. A claim about forensic cue retention would require a separate measurement that varies independently of classifier output.

7.4 Evidence needed to extend the claim

The conference reviewers raised related questions about broader robustness and repeatability. The journal makes progress by separating training initialization from test-source resampling, documenting how the operating rule interacts with class-specific effects, and retaining source links through the analysis. These changes address inference and diagnosis within the registered scope. They leave broader shifts unresolved. The locked within-release evaluation extends the positive stress-ranking contrast to new speakers and a different attack set. External-corpus, physical-channel, and multi-architecture transfer remain open.

The held-out ranking differences are smaller than the development differences, while the pooled FAKE miss reductions are larger. The different populations and denominators preclude a controlled estimate of the effect of changing partition. The A05 and A06 development errors cannot be projected directly onto A07 through A19. The official corpus documentation relates some attack algorithms across partitions, and the five-column metadata does not identify full generator lineage. The scientific conclusion should therefore be phrased in terms of the designated partitions and conditions. Claims of genuinely unseen generator robustness require an additional design with independently verified generator provenance.

7.5 What the held-out evidence changes

The positive stressed ranking contrast recurs in both held-out blocks, and pooled stressed FAKE misses decrease. The result supports transfer of those directions under the locked within-release design. Several operating-point details change. Both models trained with channel views have fewer clean REAL false alarms on held-out, whereas the development B effect reversed between seeds. Stress REAL false alarms are nearly tied in held-out 9107 and substantially lower in 9207. These observations constrain any claim that augmentation systematically shifts errors from one class to the other.

The shared adverse attack pattern is more specific. A17–A19 account for additional stressed misses in both held-out blocks despite pooled improvements. Their aggregate increases are larger in 9207, and the intervals in that block exclude zero. This supports reporting attack-conditioned effects alongside the overall result. The attack-balanced sample assigns 64 sources to every FAKE attack, so pooled misses estimate performance under that explicit mixture. No deployment mixture was measured.

Reversal of noise and codec order retains a class-dependent effect on threshold decisions. Channel training reduces many exclusive order errors, while remaining errors and clean sensitivity losses show that score stability alone is an incomplete target. The same-source comparisons support diagnosis of the implemented operators. They leave the independent forensic mechanism unresolved.

The held-out subset was fixed before inference and all four development-selected models were scored without modifying thresholds, inputs, or source quotas. This preserves the planned comparison. The two initialization blocks and the single model family still limit inference about other fitting conditions. The study establishes empirical effects for these recorded inputs and processing settings, with broader transport requiring separate experiments.

8 Limitations and Threats to Validity

The 640-source cohort is a compute-bounded training design. It does not represent full-foundation training. A and B have limited speaker support, and attack groups have 18 or 19 sources. The bootstrap describes uncertainty under speaker resampling on these fitted models and this cohort. Related conditions within a source are dependent. Secondary subgroup diagnostics are descriptive and no multiplicity adjustment is implied.

The two initialization blocks share the same development data and checkpoint rule. The same partition supports checkpoint selection and the reported development analysis. The results are sensitive to the fixed decision threshold and selected checkpoint. In particular, guard failure applies to the selected checkpoint comparison. It does not imply that every epoch or threshold would fail.

Only one architecture was retrained in the matched design. Signal preprocessing, noise scaling, and codec response can change the score contrast independently of forensic cue retention. The current experiments provide no verified independent forensic target. The completed official evaluation is within the same corpus release, and some evaluation attacks reuse algorithms or frameworks. Cross-corpus, cross-architecture, and deployment robustness remain open.

8.1 Internal and external validity

The signal intervention controls the configured processing and uses matched inputs, but the clean cohort determines which speakers and synthesis examples can contribute. Within release, reuse of attack families prevents a simple unseen-generator interpretation. Further, the detector selects checkpoints on clean development BCE, which can couple the reported clean performance to the selection data. The registered selection prevents retrospective adjustment to the stress effects while leaving this dependence intact. Effects on other operating points and threshold policies would need a separate locked design.

The source is the unit for linking ordered views, and the speaker is the resampling unit. A model sees correlated transformed versions of the same speech signal at evaluation. A larger view count does not replace additional speakers. In particular, attack-level intervals depend on a limited number of source and speaker clusters. A different bootstrap specification, broader cohort, or leave-one-speaker-out analysis could change interval width. The current inference is limited to the reported conditional comparisons.

8.2 Engineering scope

Execution integrity, deterministic replay checks and sealed artifacts support reproducibility of the fitted comparisons. They do not establish that a production implementation has the same numerical or latency properties. The measurements use a CPU with the registered environment and include training and validation work. They provide no evidence about mobile latency, energy, streaming, or adversarial adaptation. The completed evaluation uses a bounded, attack-balanced within-release subset. Representative codecs, networks, devices, languages, and authentic speech conditions require separate measurements.

8.3 Construct validity and paired uncertainty

A FAKE score is a classifier output. It can change when the transformation affects spectral structure, temporal texture, signal scale or the preprocessing input range. Without an independently defined cue and intervention, the change cannot distinguish altered evidence from altered sensitivity to the same evidence. The matched components provide a check against an overly simple additive interpretation, while the one-block retrospective scope limits replication.

The within-seed paired estimates reflect a common test population and depend on the registered speaker clusters. The selected clean checkpoint is another source of uncertainty because different realizations of the training corpus or selection data could select another epoch. The observed differences across two blocks are useful robustness diagnostics but cannot estimate a reliable distribution over seeds, architectures or corpora. The manuscript therefore reports the two contrasts separately and treats their average as descriptive.

The held-out sample contains 64 sources per FAKE attack and 256 REAL sources from 67 speakers. This quota design gives explicit support to every registered attack and changes their weighting relative to the complete official partition. Attack-level intervals describe small correlated subgroups and remain descriptive. The held-out findings are limited to the first locked test of these models. Further use of this subset for model development would make subsequent comparisons exploratory.

9 Conclusion

Channel augmentation improved stressed ranking and reduced pooled stressed FAKE misses in both matched initialization blocks on development and on the locked within-release evaluation. The held-out results also exposed a cost in clean FAKE sensitivity and additional stressed errors on A17–A19. The classwise error pattern depended on seed, condition, and partition.

The matched training comparison, source-linked predictions, and paired speaker-cluster analysis make those differences inspectable. Reversing noise and codec order continued to produce different REAL and FAKE error transitions on the held-out sources. These findings support evaluation of lightweight countermeasures with ranking, classwise risk, attack support, and same-source channel diagnostics reported together. Broader generalization across corpora and model families, and the forensic mechanism underlying the response, remain unresolved.

Reproducibility

The experiment registry, per-source scores, checkpoint seals, and paired speaker-bootstrap outputs document the development comparisons. The figure-generation code checks source identities and recorded hashes before aggregating predictions. Section 5 describes the cohort and the scope of the evaluation.

Acknowledgment

This research was supported by the University of Economics Ho Chi Minh City (UEH), Vietnam.

References

[1] H. Tak, J. Patino, M. Todisco, et al., "End-to-end anti-spoofing with RawNet2," in Proc. *IEEE International Conference on Acoustics, Speech and Signal Processing* (2021), doi:10.1109/ICASSP39728.2021.9414234, https://arxiv.org/abs/2011.01108.

[2] J. weon Jung, H.-S. Heo, H. Tak, et al., "AASIST: Audio Anti-Spoofing using Integrated Spectro-Temporal Graph Attention Networks," in Proc. *IEEE International Conference on Acoustics, Speech and Signal Processing*, pp. 6367–6371 (2022), doi:10.1109/ICASSP43922.2022.9747766, https://arxiv.org/abs/2110.01200.

[3] A. Howard, M. Sandler, G. Chu, et al., "Searching for MobileNetV3," in Proc. *IEEE/CVF International Conference on Computer Vision* (2019), https://arxiv.org/abs/1905.02244.

[4] L. Pham, D. Tran, P. Lam, et al., "DIN-CTS: Low-Complexity Depthwise-Inception Neural Network with Contrastive Training Strategy for Deepfake Speech Detection," in Proc. *European Signal Processing Conference*, pp. 556–560 (2025), https://eurasip.org/Proceedings/Eusipco/Eusipco2025/pdfs/0000556.pdf.

[5] X. Wang, J. Yamagishi, M. Todisco, et al., "ASVspoof 2019: A large-scale public database of synthesized, converted and replayed speech," *Computer Speech & Language*, vol. 64, p. 101114 (2020), doi:10.1016/j.csl.2020.101114, https://arxiv.org/abs/1911.01601.

[6] J. Yamagishi, X. Wang, M. Todisco, et al., "ASVspoof 2021: accelerating progress in spoofed and deepfake speech detection," in Proc. *ASVspoof Workshop* (2021), doi:10.21437/ASVSPOOF.2021-8, https://arxiv.org/abs/2109.00537.

[7] X. Wang, H. Delgado, H. Tak, et al., "ASVspoof 5: Design, Collection and Validation of Resources for Spoofing, Deepfake, and Adversarial Attack Detection Using Crowdsourced Speech," *Computer Speech & Language*, vol. 95, p. 101825 (2026), doi:10.1016/j.csl.2025.101825, https://www.eurecom.fr/publication/8080/download/sec-publi-8080.0-s0885230825000506-main.pdf.

[8] N. M. Müller, P. Czempin, F. Dieckmann, A. Froghyar, and K. Böttinger, "Does Audio Deepfake Detection Generalize?" in Proc. *Interspeech*, pp. 2783–2787 (2022), doi:10.21437/Interspeech.2022-108, https://www.isca-archive.org/interspeech_2022/muller22_interspeech.pdf.

[9] N. M. Müller, P. Kawa, W. H. Choong, et al., "MLAAD: The Multi-Language Audio Anti-Spoofing Dataset," in Proc. *International Joint Conference on Neural Networks*, pp. 1–7 (2024), doi:10.1109/IJCNN60899.2024.10650962, https://www.resemble.ai/resources/mlaad-the-multi-language-audio-anti-spoofing-dataset.

[10] J.-M. Valin, K. Vos, and T. B. Terriberry, "Definition of the Opus Audio Codec," (2012), doi:10.17487/RFC6716, https://www.rfc-editor.org/rfc/rfc6716, rFC 6716.

[11] H. Maltby, J. Wall, C. Glackin, et al., "Robustness in Deepfake Speech Detection: A Survey of Failure Mechanisms Including an Experimental Case Study," *Computer Speech &*

Language, vol. 102, p. 102037 (2026), doi:10.1016/j.csl.2026.102037, https://repository.uwl.ac.uk/id/eprint/15273/.

[12] H. Tak, M. Kamble, J. Patino, M. Todisco, and N. Evans, "RawBoost: A Raw Data Boosting and Augmentation Method applied to Automatic Speaker Verification Anti-Spoofing," in Proc. *IEEE International Conference on Acoustics, Speech and Signal Processing*, pp. 6382–6386 (2022), doi:10.1109/ICASSP43922.2022.9746213, https://arxiv.org/abs/2111.04433.

[13] Z. Zhang, W. Hao, A. Sankoh, et al., "I Can Hear You: Selective Robust Training for Deepfake Audio Detection," in Proc. *International Conference on Learning Representations* (2025), https://www.cs.columbia.edu/~junfeng/papers/fsat/.

[14] S. Sagawa, P. W. Koh, T. B. Hashimoto, and P. Liang, "Distributionally Robust Neural Networks for Group Shifts: On the Importance of Regularization for Worst-Case Generalization," in Proc. *International Conference on Learning Representations* (2020), https://arxiv.org/abs/1911.08731.

[15] T. M. Wani and I. Amerini, "Dynamic Knowledge Condensation with Audio-Selective Transformer for Audio Deepfake Detection," *Discover Computing*, vol. 28, p. 231 (2025), doi:10.1007/s10791-025-09746-4, https://link.springer.com/article/10.1007/s10791-025-09746-4.

[16] I. N. Hidekel, G. Lifshitz, K. Cohen, and D. Raviv, "SONAR: Spectral-Contrastive Audio Residuals for Generalizable Deepfake Detection," in Proc. *International Conference on Machine Learning*, vol. 306, pp. 42998–43012 (2026), https://proceedings.mlr.press/v306/hidekel26a.html.

[17] H. Shi, X. Shi, S. Dogan, et al., "Benchmarking Audio Deepfake Detection Robustness in Real-World Communication Scenarios," in Proc. *European Signal Processing Conference*, pp. 566–570 (2025), doi:10.23919/EUSIPCO63237.2025.11226601, https://eusipco2025.org/wp-content/uploads/pdfs/0000566.pdf.

[18] C. Y. Kwok, J. Q. Yip, Z. Qiu, C. H. Chi, and K. Y. Lam, "Bona Fide Cross Testing Reveals Weak Spot in Audio Deepfake Detection Systems," in Proc. *Interspeech*, pp. 2230–2234 (2025), doi:10.21437/Interspeech.2025-172, https://www.isca-archive.org/interspeech_2025/kwok25_interspeech.html.

[19] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On Calibration of Modern Neural Networks," in Proc. *International Conference on Machine Learning*, vol. 70, pp. 1321–1330 (2017), https://proceedings.mlr.press/v70/guo17a.html.

[20] Y. Geifman and R. El-Yaniv, "SelectiveNet: A Deep Neural Network with an Integrated Reject Option," in Proc. *International Conference on Machine Learning*, vol. 97, pp. 2151–2159 (2019), https://proceedings.mlr.press/v97/geifman19a.html.

[21] O. Pascu, A. Stan, D. Oneata, E. Oneata, and H. Cucu, "Towards Generalisable and Calibrated Audio Deepfake Detection with Self-Supervised Representations," in Proc. *Interspeech*,

pp. 4828–4832 (2024), doi:10.21437/Interspeech.2024-1302, https://www.isca-archive.org/interspeech_2024/pascu24_interspeech.html.

[22] H. Tak, M. Todisco, X. Wang, et al., "Automatic Speaker Verification Spoofing and Deepfake Detection Using Wav2vec 2.0 and Data Augmentation," in Proc. *The Speaker and Language Recognition Workshop (Odyssey 2022)*, pp. 112–119 (2022), doi:10.21437/Odyssey.2022-16, https://www.isca-archive.org/odyssey_2022/tak22_odyssey.html.

[23] I.-P. Ciobanu, A.-I. Hîji, N. C. Ristea, et al., "XMAD-Bench: Cross-Domain Multilingual Audio Deepfake Benchmark," in Proc. *Findings of the Association for Computational Linguistics: EACL 2026*, pp. 3109–3120 (2026), doi:10.18653/v1/2026.findings-eacl.162, https://aclanthology.org/2026.findings-eacl.162/.

[24] J. A. Hanley and B. J. McNeil, "The Meaning and Use of the Area Under a Receiver Operating Characteristic (ROC) Curve," *Radiology*, vol. 143, pp. 29–36 (1982), doi:10.1148/radiology.143.1.7063747, https://doi.org/10.1148/radiology.143.1.7063747.

[25] D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," in Proc. *International Conference on Learning Representations* (2015), https://arxiv.org/abs/1412.6980.

[26] C. A. Field and A. H. Welsh, "Bootstrapping Clustered Data," *Journal of the Royal Statistical Society: Series B*, vol. 69, pp. 369–390 (2007), doi:10.1111/j.1467-9868.2007.00593.x, https://rss.onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-9868.2007.00593.x.